



SIMULTANEOUS LINEAR EQUATIONS

Class 12 Applied Maths — Fast MCQ Tricks — By Thunderstudy

Think of equations like two roads crossing. The solution = where both roads meet. Most MCQs don't need full solving — just smart checking!

✓ 1 The CROSS-MULTIPLICATION MAGIC

Given: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

⚡ **Direct Formula (No Long Steps):**

$$x = (b_1c_2 - b_2c_1) / (a_1b_2 - a_2b_1)$$
$$y = (c_1a_2 - c_2a_1) / (a_1b_2 - a_2b_1)$$

Memory Trick: Write in circle order $x \rightarrow b \ c$, $y \rightarrow c \ a$, den. $\rightarrow a \ b$. Rotate letters clockwise .

✓ 2 DETERMINANT TEST (MCQ TRICK)

Calculate: $D = a_1b_2 - a_2b_1$

Condition	Meaning
$D \neq 0$	Unique solution ✓
$D = 0$ (Ratios equal)	Infinite solutions ∞
$D = 0$ (Ratios unequal)	No solution ✗

🧠 **Ratio Trick:**

Check a_1/a_2 , b_1/b_2 , c_1/c_2 . All equal $\rightarrow$ Same line. First two equal, third different $\rightarrow$ Parallel lines.

✓ 3 **OPTION CHECKING METHOD (TOPPER SECRET 🤔)**

MCQ gives options? DON'T solve. Put options (x,y) into BOTH equations. If both satisfy $\rightarrow$ correct answer. This is usually the fastest method in the exam!

✓ 4 ADD-SUBTRACT SPEED TRICK

If coefficients look similar (e.g., $3x+2y=10$ and $3x-2y=2$), just add them! $6x=12 \rightarrow x=2$. Instant solution 😎

✓ 5 Zero Coefficient Shortcut

If numbers look messy, try making one variable 0 mentally. It helps eliminate options fast.

🚀 EXAM SPEED STRATEGY (Golden Rule)

Question Type	Best Method
Find solution	Cross multiplication
Nature of solution	Determinant / Ratio Test
Options given	Substitute option
Similar coefficients	Add/Subtract

🧠 10-Second Memory Formula

- 👉 D decides destiny
- 👉 Options beat algebra
- 👉 Add when signs differ
- 👉 Cross multiply for direct answer

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